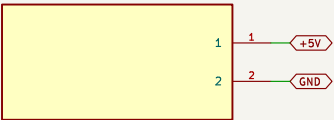


1	2	3	4	5	6	7	8
COLLOQUY OF MOBILES / CONTROL PCB V2 Revision A prototype Fixed harness / Mega2560 / U2D2 / five complete audio channels							

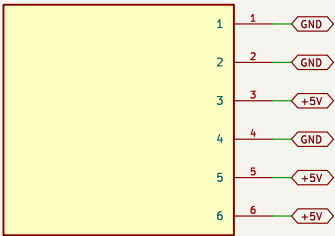
Colloquy of Mobiles		
Sheet: /		
File: colloquy-control-v2.kicad_sch		
Title: Colloquy control v2: system overview		
Size: A3	Date: 2026-09-29	Rev: A-prototype
KiCad E.D.A. 9.0.7	Id: 1/10	

Power, reference bond and service points

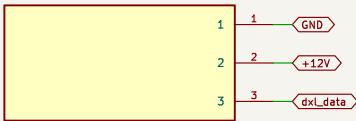
J2
DC jack



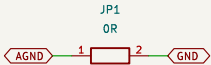
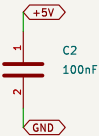
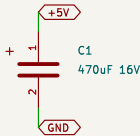
J6
screw bridge



J7
JST EH 3



J2: center-positive REGULATED 5V ONLY.
J6 is a parallel power landing; same-polarity lugs are commoned.
J7 pin2 supplies 12V servo distribution; it never powers the Mega.



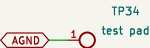
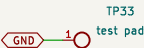
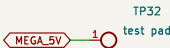
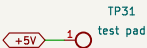
M1
U2D2



JP1: single AGND-to-GND bond beside Mega.
M1 is the existing U2D2 mechanical mount; connect TTL lead via J7.
External PSU must provide current limiting / branch protection.



RS1 uses MEGA_5V; D2 is reserved; there is no five-body mute harness.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /Power, reference bond and service points/
File: power.kicad_sch

Title: Power, reference bond and service points

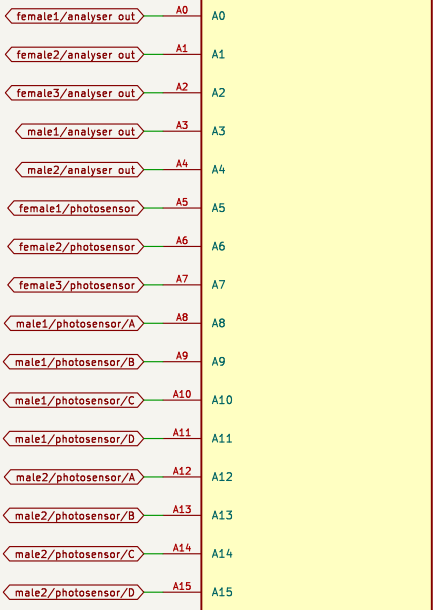
Size: A3	Date: 2026-09-29	Rev: A-prototype
KiCad E.D.A. 9.0.7		Id: 2/10

Arduino Mega 2560: firmware 4 pin assignment

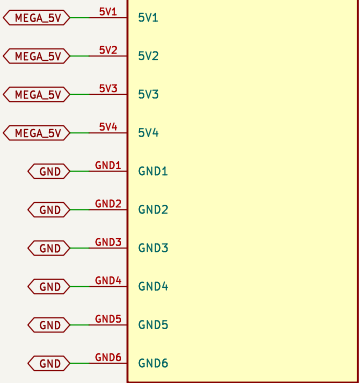
A1A
Arduino Mega 2560



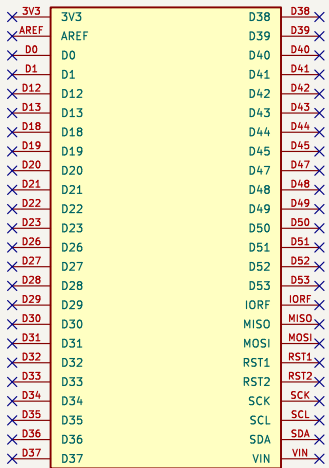
A1B
Arduino Mega 2560



A1C
Arduino Mega 2560



A1D
Arduino Mega 2560



USB powers the Mega. External +5V does not join MEGA_5V.
All physical 5V and GND shield pads are explicitly assigned.
D0/D1, D13, D20/D21 remain unused by this PCB.

Prototype: verify harness polarity and sensor values before connection

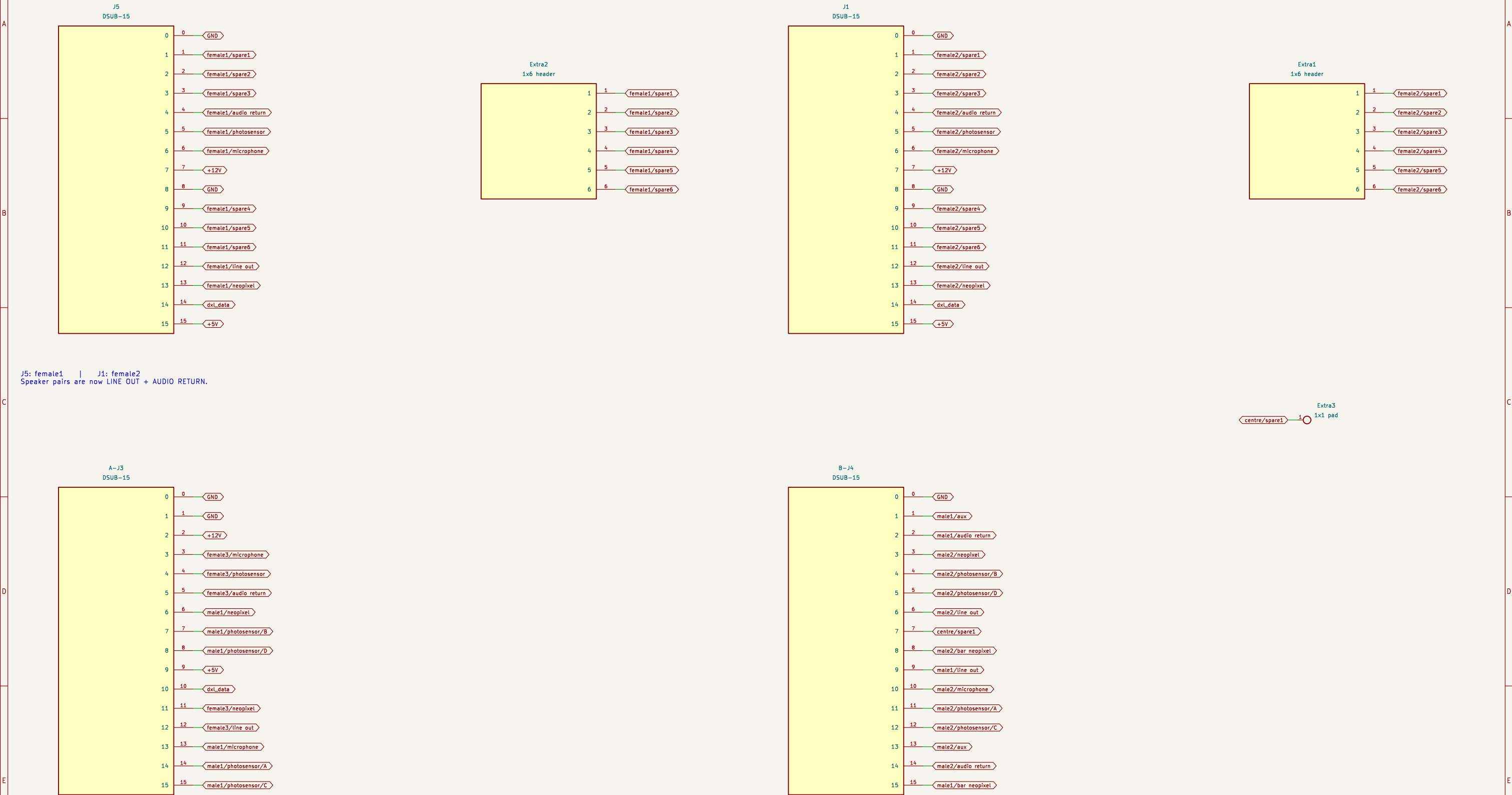
Colloquy of Mobiles

Sheet: /Arduino Mega 2560: firmware 4 pin assignment/
File: controller.kicad_sch

Title: Arduino Mega 2560: firmware 4 pin assignment

Size: A3	Date: 2026-09-29	Rev: A-prototype
KiCad E.D.A. 9.0.7		Id: 3/10

Fixed enclosure connectors and spare conductors



J5: female1 | J1: female2
Speaker pairs are now LINE OUT + AUDIO RETURN.

A-J3: female3 + shared center power + male1 sensors
B-J4: male1/male2 signals; NO POWER on this connector.
Use J7 for Dynamixel GND / +12V / DATA. Shells join GND.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /Fixed enclosure connectors and spare conductors/
File: harness.kicad_sch

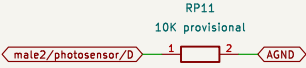
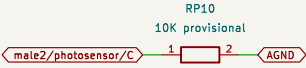
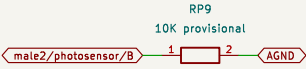
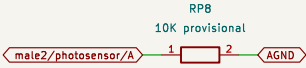
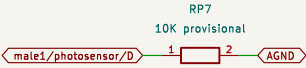
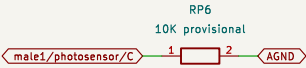
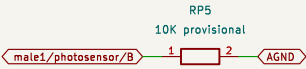
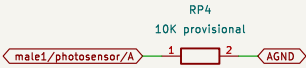
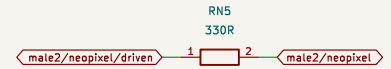
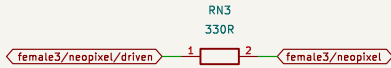
Title: Fixed enclosure connectors and spare conductors

Size: A3 Date: 2026-09-29 Rev: A-prototype
KiCad E.D.A. 9.0.7 Id: 4/10

Pixel damping, spare controls and photosensor dividers

330R series at the controller end

10K is an assembly assumption: calibrate each sensor



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /Pixel damping, spare controls and photosensor dividers/

File: io.kicad_sch

Title: Pixel damping, spare controls and photosensor dividers

Size: A3

Date: 2026-09-29

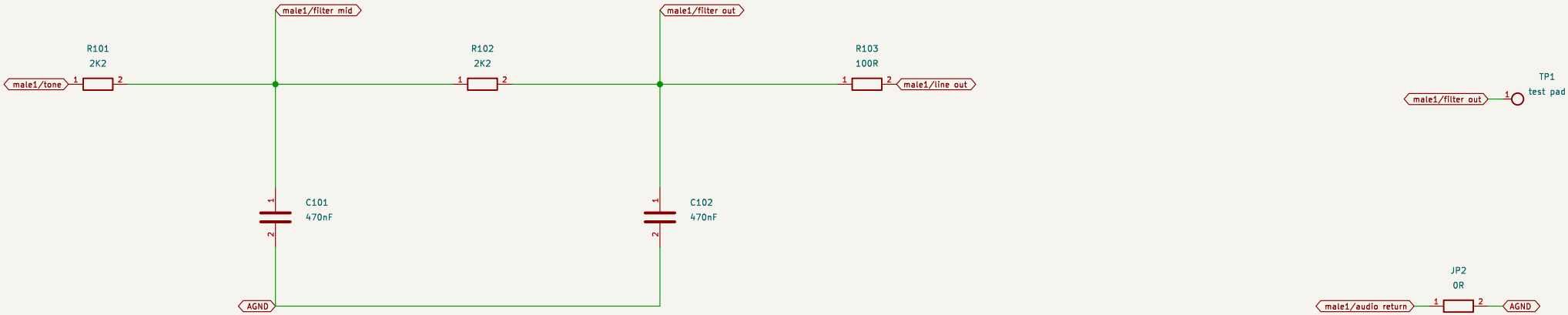
Rev: A-prototype

KiCad E.D.A. 9.0.7

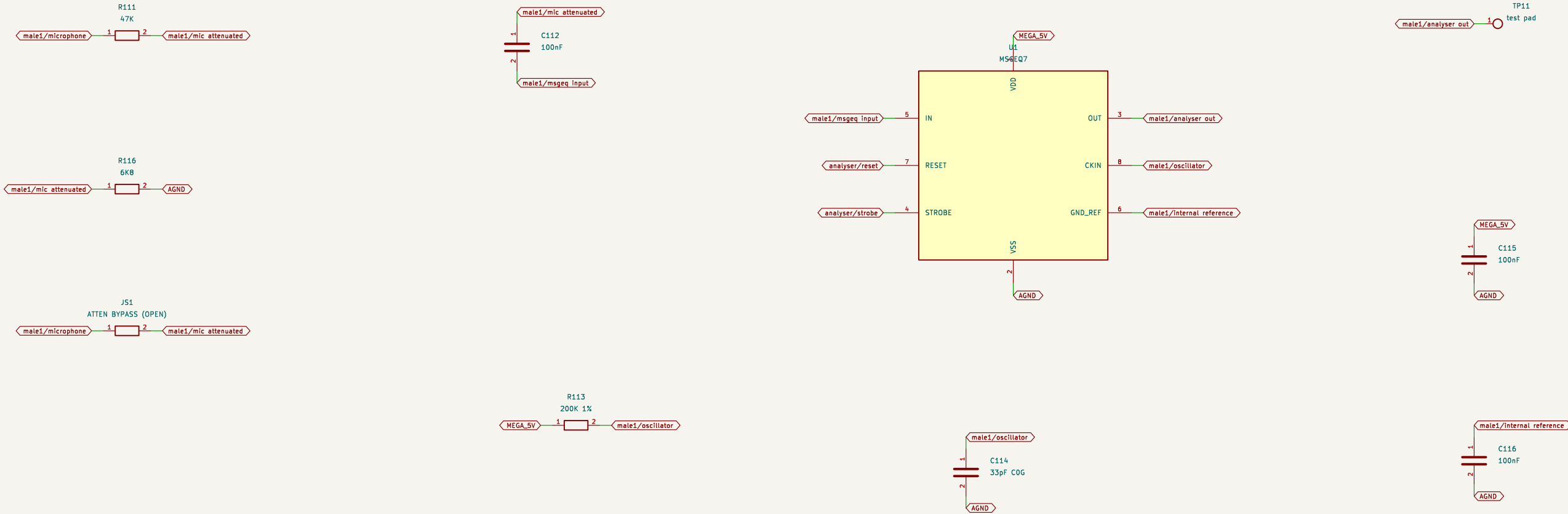
Id: 5/10

male1: 160 Hz voice / A3 receiver

TRANSMIT D11 / 160 Hz hardware timer -> two RC stages -> line-level harness



RECEIVE male1 microphone -> 0.126 attenuation -> AC coupling -> MSGEQ7 -> A3



JS default OPEN. Bridge only after measuring input headroom.
R=47K/6K8: 2Vpp microphone becomes about 0.25Vpp at IN.
Pin6 is an INTERNAL 2.5V REFERENCE: bypass, never short to GND.

Oscillator: 200K / 33pF incl. stray C.
Keep clock and bypass parts at DIP pins.
Audio return bonds at the local OR link.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /male1: 160 Hz voice / A3 receiver/

File: audio1.kicad_sch

Title: male1: 160 Hz voice / A3 receiver

Size: A3

Date: 2026-09-29

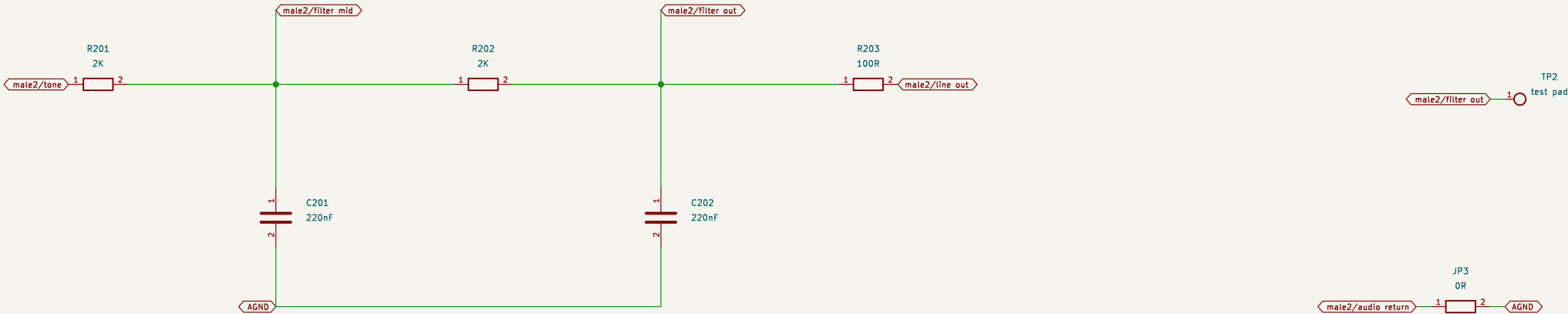
Rev: A-prototype

KiCad E.D.A. 9.0.7

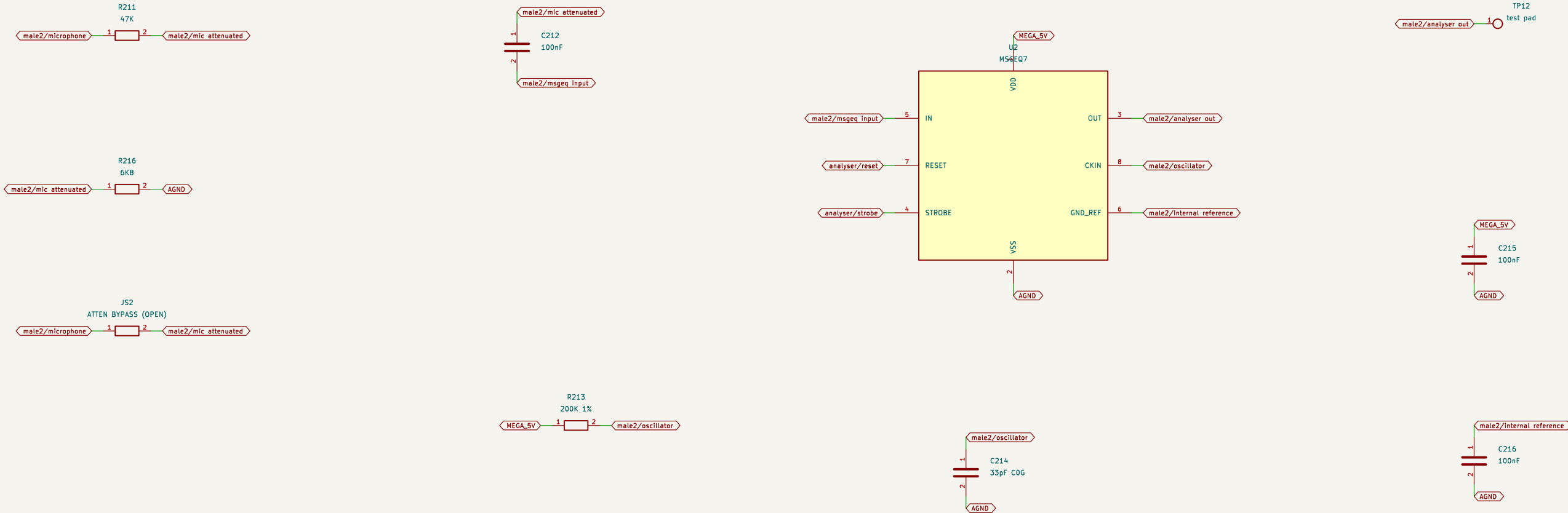
Id: 6/10

male2: 400 Hz voice / A4 receiver

TRANSMIT D5 / 400 Hz hardware timer -> two RC stages -> line-level harness



RECEIVE male2 microphone -> 0.126 attenuation -> AC coupling -> MSGEQ7 -> A4



JS default OPEN. Bridge only after measuring input headroom.
R=47K/6K8: 2Vpp microphone becomes about 0.25vpp at IN.
Pin6 is an INTERNAL 2.5V REFERENCE: bypass, never short to GND.

Oscillator: 200K / 33pF incl. stray C.
Keep clock and bypass parts at DIP pins.
Audio return bonds at the local OR link.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /male2: 400 Hz voice / A4 receiver/

File: audio2.kicad_sch

Title: male2: 400 Hz voice / A4 receiver

Size: A3

Date: 2026-09-29

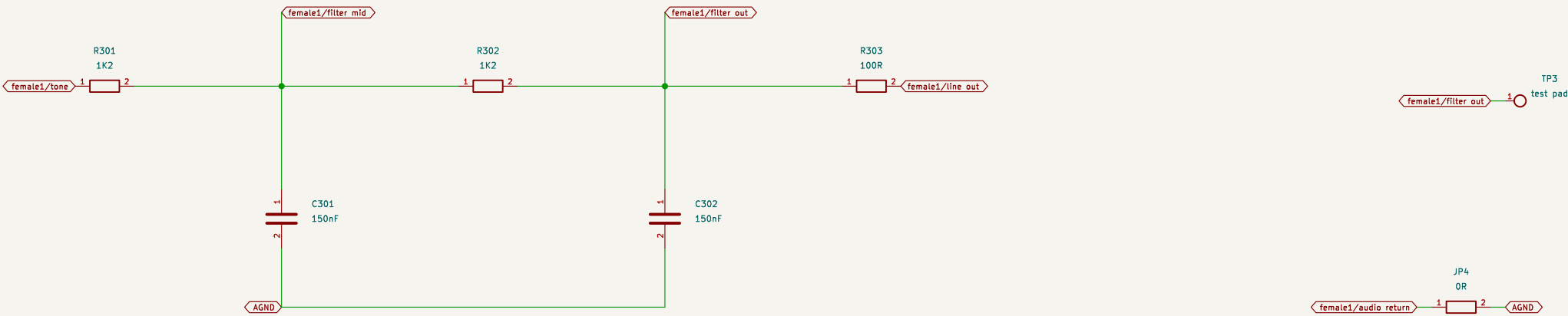
Rev: A-prototype

KiCad E.D.A. 9.0.7

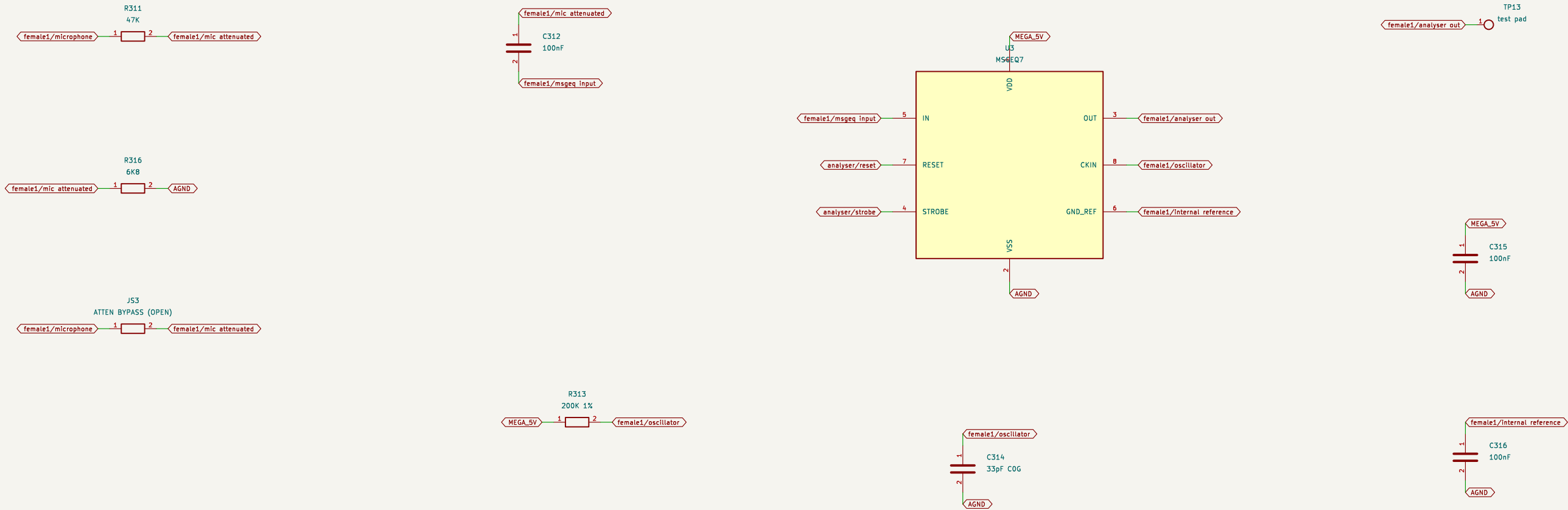
Id: 7/10

female1: 1000 Hz voice / A0 receiver

TRANSMIT D6 / 1000 Hz hardware timer -> two RC stages -> line-level harness



RECEIVE female1 microphone -> 0.126 attenuation -> AC coupling -> MSGEQ7 -> A0



JS default OPEN. Bridge only after measuring input headroom.
R=47K/6K8: 2Vpp microphone becomes about 0.25Vpp at IN.
Pin6 is an INTERNAL 2.5V REFERENCE: bypass, never short to GND.

Oscillator: 200K / 33pF incl. stray C.
Keep clock and bypass parts at DIP pins.
Audio return bonds at the local OR link.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /female1: 1000 Hz voice / A0 receiver/

File: audio3.kicad_sch

Title: female1: 1000 Hz voice / A0 receiver

Size: A3

Date: 2026-09-29

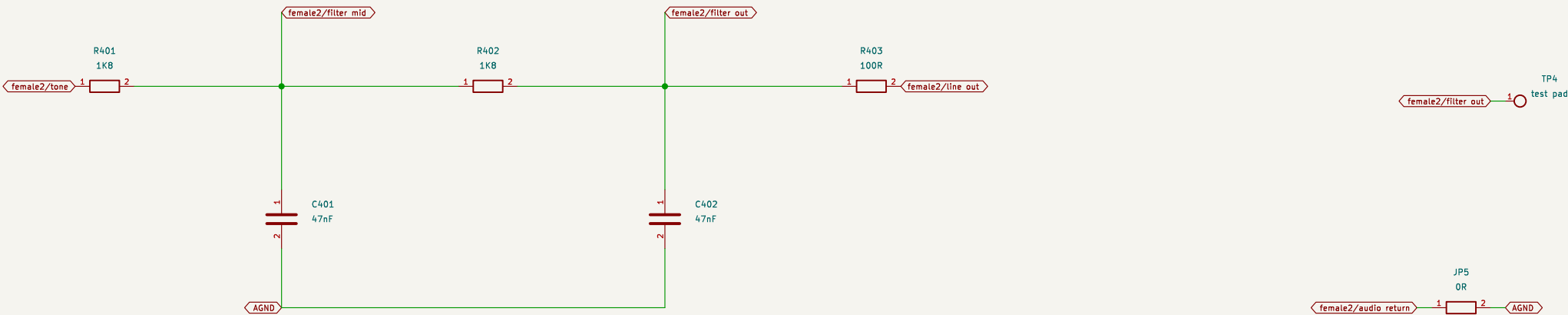
Rev: A-prototype

KiCad E.D.A. 9.0.7

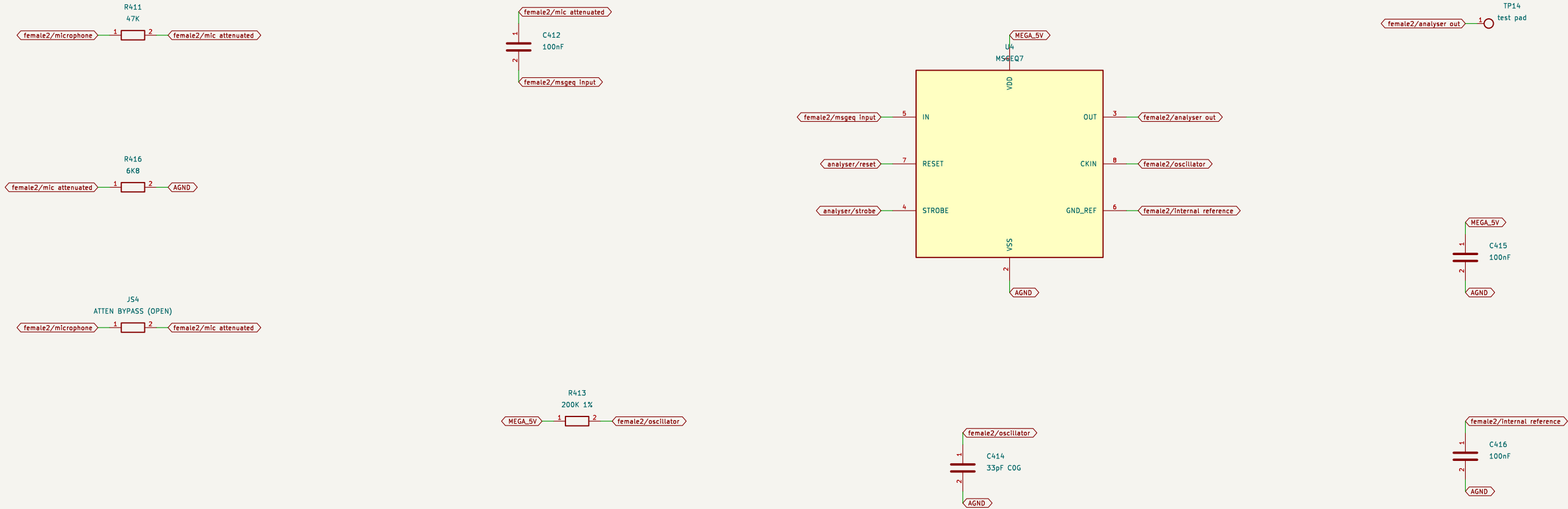
Id: 8/10

female2: 2500 Hz voice / A1 receiver

TRANSMIT D46 / 2500 Hz hardware timer -> two RC stages -> line-level harness



RECEIVE female2 microphone -> 0.126 attenuation -> AC coupling -> MSGEQ7 -> A1



JS default OPEN. Bridge only after measuring input headroom.
R=47K/6K8: 2Vpp microphone becomes about 0.25Vpp at IN.
Pin6 is an INTERNAL 2.5V REFERENCE: bypass, never short to GND.

Oscillator: 200K / 33pF incl. stray C.
Keep clock and bypass parts at DIP pins.
Audio return bonds at the local OR link.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /female2: 2500 Hz voice / A1 receiver/

File: audio4.kicad_sch

Title: female2: 2500 Hz voice / A1 receiver

Size: A3

Date: 2026-09-29

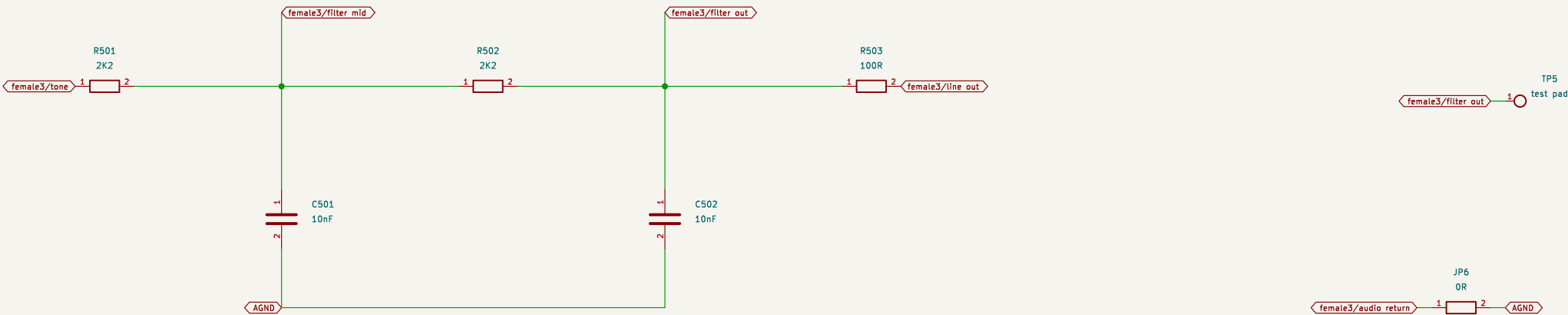
Rev: A-prototype

KiCad E.D.A. 9.0.7

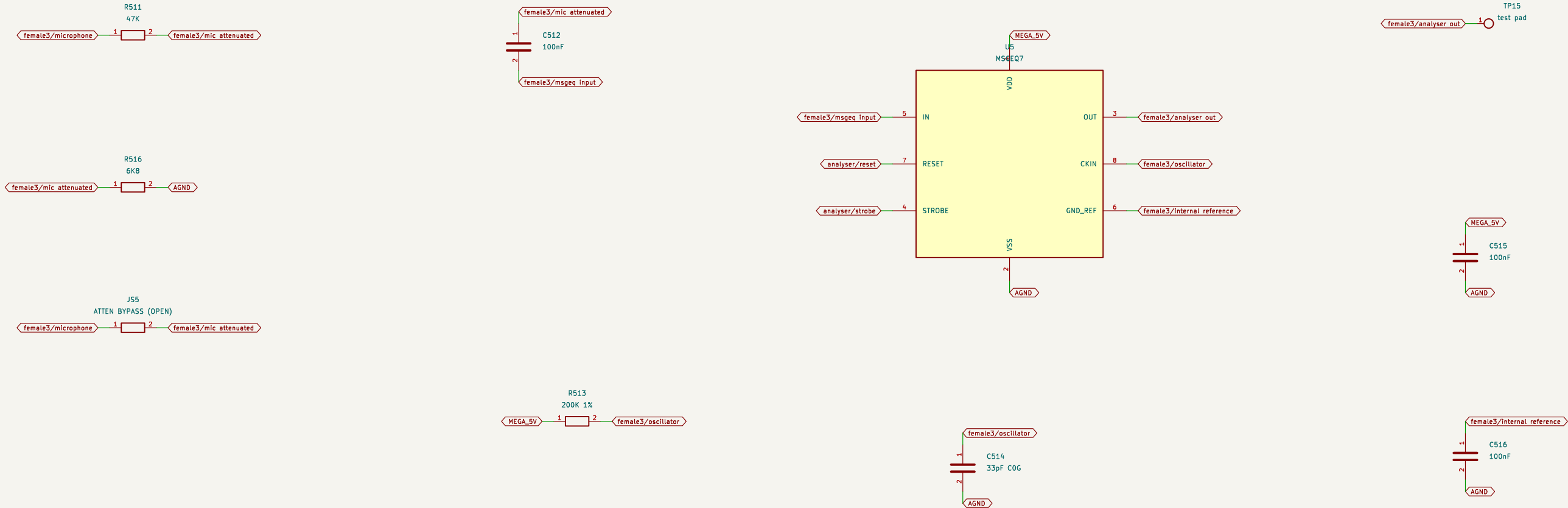
Id: 9/10

female3: 6250 Hz voice / A2 receiver

TRANSMIT D10 / 6250 Hz hardware timer -> two RC stages -> line-level harness



RECEIVE female3 microphone -> 0.126 attenuation -> AC coupling -> MSGEQ7 -> A2



JS default OPEN. Bridge only after measuring input headroom.
R=47K/6K8: 2Vpp microphone becomes about 0.25Vpp at IN.
Pin6 is an INTERNAL 2.5V REFERENCE: bypass, never short to GND.

Oscillator: 200K / 33pF incl. stray C.
Keep clock and bypass parts at DIP pins.
Audio return bonds at the local OR link.

Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /female3: 6250 Hz voice / A2 receiver/

File: audio5.kicad_sch

Title: female3: 6250 Hz voice / A2 receiver

Size: A3

Date: 2026-09-29

Rev: A-prototype

KiCad E.D.A. 9.0.7

Id: 10/10